Personalized Checkout A/B Test

Executive Memo — Results & Recommendation

Decision

Roll out the personalized checkout carousel to frequent shoppers. Hold off on occasional shoppers pending UX refinement.

Overview

A/B test of a personalized "Last-Minute Additions" checkout carousel against a standard checkout flow, run on 13,059 assigned users (of 20,000 total). Assignment integrity was verified via a Chi-Square SRM test (p = 0.6554) - no sample ratio mismatch detected, confirming the results below are reliable.

Primary Results

Metric	Control	Treatment	Lift	p-value	Result
AOV	$46.41	$53.56	+15.4%	0.0005	Significant
UPT	3.48	4.08	+17.1%	<0.0001	Significant
RPRU	$85.63	$97.72	+14.1%	0.0023	Significant
Abandonment	22.18%	24.91%	+12.3% rel.	<0.0001	Significant (guardrail)

The carousel produced a significant increase in spending across every primary metric. It also produced a significant increase in cart abandonment (22.18% to 24.91%), the trial's guardrail metric, meaning the feature drives more spend, but at the cost of some additional checkout friction.

Segment Findings

The effect is not uniform. Frequent shoppers (3+ orders in the prior 90 days) showed a stronger spending response and a smaller abandonment cost than occasional shoppers:

Segment	AOV Lift	RPRU Lift	Abandonment Increase	p-values
Frequent	+19.3%	+18.2%	+9.9% rel.	0.010 / 0.021 / 0.023
Occasional	+13.4%	+12.0%	+13.2% rel.	0.014 / 0.034 / <0.0001

Occasional shoppers saw a smaller lift and a larger relative increase in abandonment (+13.2% vs. +9.9%), suggesting the carousel's personalization is more credible and less disruptive for users the app already has purchase history on.

Revenue Impact

- Full rollout (all users): +$7.3M projected annual incremental revenue but carries the higher abandonment cost from occasional shoppers.
- Frequent-shoppers-only rollout (~20.2% of user base): +$3.6M projected annual incremental revenue, at a lower abandonment cost.

Figures reflect the incremental effect of the tested feature only, based on the observed treatment-vs-control spend difference during the experiment window.

Recommendation

- Deploy the carousel to frequent shoppers now — this segment shows the strongest returns and the lowest guardrail cost.
- Hold off on occasional shoppers until the checkout flow is refined (e.g., an easier skip/dismiss option) to reduce added friction.
- Monitor abandonment rate post-launch as a standing guardrail, particularly if rollout is later extended to occasional shoppers.